Effects of Cabbage (*Brassica oleracea*) Extract in Prolonging the Shelf-Life of Radish (*Raphanus sativus*)

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ABSTRACT

The main objective of this study was to extract flavonoids from cabbages, apply the formulated extracts to the radish, and measure the effects, while simultaneously repurposing the cabbages, focusing on parameters such as weight loss, signs of decay, and firmness. The study created 250 mL cabbage extracts at 20%, 40%, and 60% concentrations using microwave-assisted extraction and tested for flavonoids using Alkaline Reagent Test and Shinoda Test, which was applied to the radishes at a 7-day period. Results show that all treated groups exhibited slightly greater weight loss compared to control but was not significant with a p-value of 0.915. All treated groups showed a reduction in decay signs compared to control, but the difference lacked statistical significance with a p-value of 0.958. All treated groups showed significantly improved firmness with a p-value of 0.017, with a post-hoc analysis that revealed a significant firmness improvement with 20% concentration. While the cabbage extract did not significantly impact weight loss or signs of decay of radishes compared to the control, all treated groups exhibited significantly firmer radishes, with the 20% concentration showing the most efficacy. Further studies could explore alternate samples and extractions and modified controlled storage.

Keywords: *Cabbage extract, Flavonoids, Microwave-assisted extraction, Shelf-life*

INTRODUCTION

For the past several years, society all over the world has experienced the issue of food waste at some point. This led to the unfathomable level of hunger in areas under poverty. Including the Philippines, which had generated over 930 million tons of food waste in 2019. A statistic from DOST-FNRI (2023) mentions that the city of Manila has the highest rate of food waste in the country.
As mentioned by Sanchez (2019), approximately 42% of the vegetables in the Philippines that are wasted pose a challenge to the country's food security. According to Feiereisen (2020) and Debora (2020), spoilage in vegetables occurs due to their high moisture content.

An example of a high moisture vegetable are Radishes. It is one of the fastest vegetables to spoil within two to four days, for room temperature, as stated by Williams (2022). Radishes contain 90–95% water, which composes most of their weight and contributes to their short shelf life, as mentioned by Huynh (2019).

According to Frillman (2022), shelf life is one of the most sought-after qualities that consumers look for. Shelf-life has both extrinsic (environmental factors during storage) and intrinsic factors (inherent to the radish itself). The most relevant factors for shelf-life are extrinsic factors, specifically the application of cabbage extract at varying levels, because while intrinsic factors are less controllable, extrinsic factors offer opportunities for intervention. One possible way of extending the shelf-life is by repurposing unwanted agricultural produce like cabbages. Many studies that were mentioned by Pietta (2010) suggest that Flavonoids are rich in bioactive compounds, including antioxidant activities. Kaempferol and quercetin were the major flavonoids found, as mentioned by Chun (2014).

Coating flavonoids might interact with the skin of the radish, which prolongs its life. This study was inspired by the product called "Apeel©," in which they coat their products with a natural preservative that extends shelf life. Apeel stated the research of Currier (2017), who stated that various natural solutions to shelf-life were also used under the USDA National Organic Program.

A way to extract said flavonoids is through a non-conventional method called microwave-assisted extraction, based on electromagnetic waves that cause the cell structure to change, as mentioned by Nour (2021). Quercetin and kaempferol can either be free or glycosylated. Glycosylated flavonols can be soluble in water due to their structure.

The study aims to extract the flavonoids (quercetin and kaempferol) from unwanted cabbages and apply the formulated extract with the different concentrations to the radishes, in which the effects were compared based on the radishes weight loss, signs of decay, and firmness.

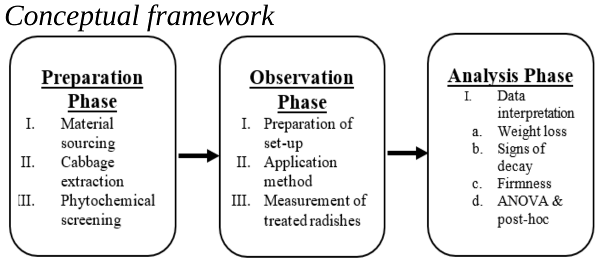
Repurposing unwanted cabbages to measure the effects on prolonging the shelf-life of radishes has produced several problems. Firstly, what are the effects of the different concentrations of cabbage extracts based on weight loss (g), signs of decay (%), and firmness (Pa)? Secondly, out of the following concentrations (20%, 40%, and 60%), which concentration of cabbage extract is most effective in prolonging the shelf-life of radishes (*Raphanus sativus*)? Lastly, is there a significant difference in the shelf-life of radish (*Raphanus sativus*) added with the different concentrations of cabbage extracts based on the quantitative parameters mentioned?

This study explores the unknown potential of using cabbage extract, as a potential natural preservative, to extend radishes' shelf life. This study might also open the possibility for future research on repurposed natural food preservation using modern techniques. The findings, regardless of the outcome, will contribute insights that would lead to a novel solution to food waste.

METHODS

The study used an experimental research design adapted from Sirisilla (2023). The study was conducted at the DLSAU LSB-711 Laboratory, in which the randomized complete block design (RCBD) was used as the sampling method of observation.

Figure 1
Conceptual framework



Material sourcing

Extraction of cabbage required a cabbage head, two liters of distilled water, a microwave oven that can reach 373 °K (99.85 °C), (3) 500mL beakers, (3) 300mL jars, (3) watch glass, glass rod, thermometer, cooler, knife and cutting board. NaOH, HCl, and magnesium metal were used for phytochemical tests. Application of cabbage extract required sixty-four (64) radishes, (12) 12L containers, a thermometer/hygrometer. An electronic scale, an analytical balance, and a fruit penetrometer were the instruments used. Radishes that were dry, firm, and had no visible decay were used.

Cabbage extraction

The study created three (3) cabbage extract concentrations of 20%, 40%, and 60%. A cabbage was cut into small pieces and distributed into 50g, 100g, and 150g portions, which were added to 250 mL of distilled water into three separate beakers. While extracting, the temperature of the extract solutions must not exceed 100°C and a time limit of 25 minutes. The researchers microwaved the solution for 45 seconds with 1 to 2 minutes to mix. The cycle continued until the 25-minute time limit.

Phytochemical screening

The cabbage extracts were screened for flavonoids using the alkaline reagent test and Shinoda test. For the alkaline reagent test, 2-3 drops of 2M NaOH solution were applied to 2mL of all cabbage extract concentrations. This resulted in a rich yellow color that fades into colorless after adding a few drops of 6M HCl. For the Shinoda Test: 1-2 shards of magnesium were added into 2mL of all cabbage extract concentrations and then followed by a drop of 6M of HCl. This resulted in a faint pink/purple color.

Preparation of set-up

Twelve containers were prepared for the radishes. Each container contained four radishes, totaling over forty-eight. Each concentration group and control had 3 containers assigned as Control (C), 20% Concentration (T_1), 40% Concentration (T_2), and 60% Concentration (T_3).

Application method

Radishes were divided into control and treatment. The control group was submerged in distilled water, while the treatment group received a cabbage extract solution at specified concentrations. All radishes were dipped in their respective solutions for 15 seconds, followed by 10 seconds for dripping the excess liquid.

Measurement of treated radishes

Initial weight in (g) that was measured using an electronic scale and initial firmness in (Pa) that was measured using a fruit penetrometer were measured before application. Radishes that were measured for firmness are to be disposed of. Final weight in (g) and firmness in (Pa) were measured on Day 7.

Data interpretation

Weight loss was calculated by subtracting the initial and final measurement of individual radishes using the formula: $WL = W_i - W_f$. Where W_i is the initial weight of a radish and W_f is the final weight measured. Signs of decay were calculated by using the postharvest decay formula: PDP = ($\frac{n}{n_t}$ x 100) based on Seun (2022), Where n is the number of decayed radishes and n_t is the total number of radishes. Signs of decay were determined by visual necrotic spots and presence of water precipitated from the radishes with foul odor.

Figure 2

Signs of decay on radishes

Firmness of the radish in all set-ups was measured using the fruit penetrometer with the unit being10^5 Pa.

After organization of all data, Jamovi version "2.3.28" was used to calculate the Welsch's One-Way ANOVA and Tukey HSD Post-hoc. The Data Set of both ANOVA are from the Weight Loss of each individual radish, post-harvest decay percentage per group and Firmness of four (4) radishes with the highest firmness per concentration group.

RESULTS

Radishes were prepped, measured, and applied by coating for 15 sec in distilled water for control or their respective cabbage extract concentration that was created using microwave-assisted extraction at around 58.3°C for 25 min. Applied radishes were dripped for 10 seconds, and was monitored for 7 days.

Weight loss of the radishes was measured using an electronic scale. Weight Loss was measured, and results are displayed as Day 0 and Day 7. The total weight loss was measured by calculating the difference by subtracting the final weight to the initial weight of the radish. Lower weight loss indicates better results.

Table 1 shows the summary of weight loss, presented as grams (g):

Table 1

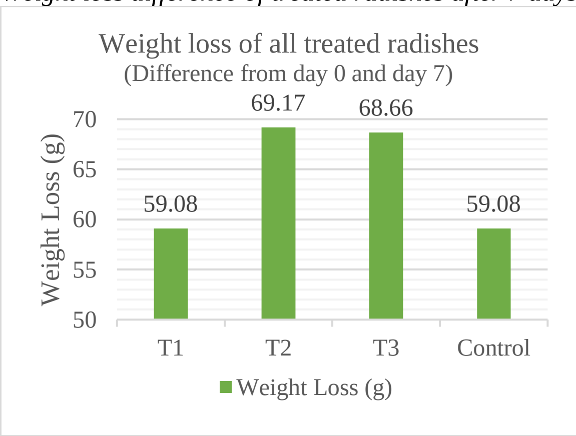
Weight loss of treated radishes at day 0 & day 7

Groups	Day 0	Day 7	Difference
T1	119.54 g	60.46 g	**59.08 g**
T2	127.88 g	58.71 g	**69.17 g**
T3	123.79 g	55.13 g	**68.66 g**
Control	121.87 g	62.79 g	**59.08 g**

Figure 3 shows the bar graph showing weight loss, presented as grams (g):

Figure 3

Weight loss difference of treated radishes after 7 days



Signs of decay of the radishes were measured using visual observations. The formula was used to get a quantified percentage of decayed radishes per subgroup. A lower percentage indicated better results.

Table 2 shows the summary of initial and final post-harvest decay percentage:

Table 2

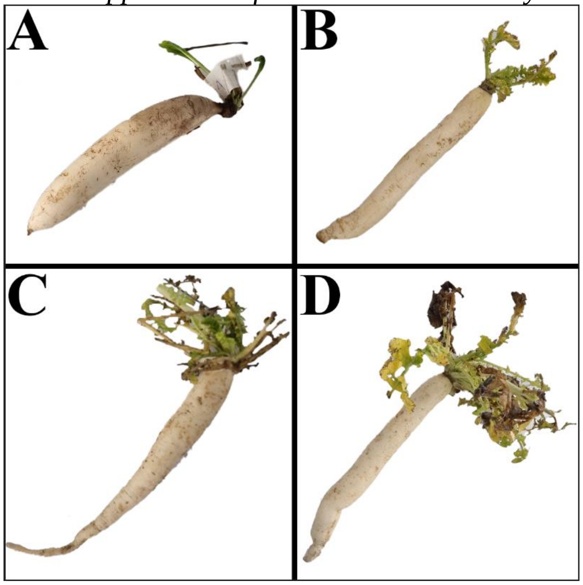
Percentage decay of treated radishes at day 0 & day 7

Groups	Day 0	Day 7
T1	0%	**66.7%**
T2	0%	75%
T3	0%	75%
Control	0%	83.3%

Figure 4 shows the signs of decay of individual radishes per group after seven (7) days of observation:

Figure 4

Radish appearance of treated radishes at day 7



Radish A was treated without cabbage extract but was coated in distilled water. Radish B was treated with 20% concentration. Radish C was treated with 40% concentration. Radish D was treated with 60% concentration. Based on visual observations, radish B seemed to be the least decayed, with no signs of decay compared to the others.

Firmness of the radishes was measured using a fruit penetrometer. 4 radishes per treatment group were used in initial measurement, and all radishes on Day 7, using the four highest measured values. Higher firmness indicated better results.

Table 3 shows the summary of the firmness readings, presented as Pascals (Pa):

Table 3

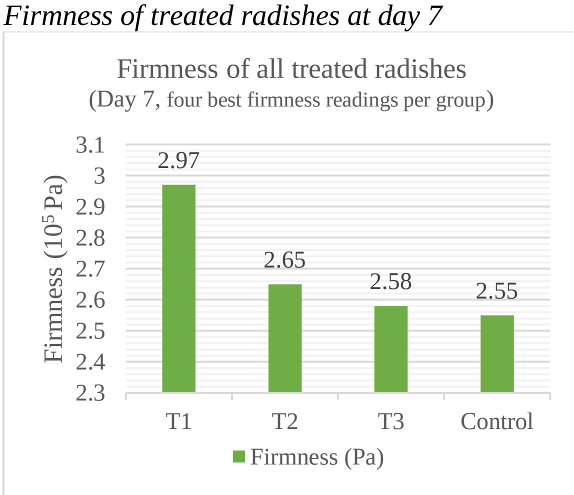
Firmness of treated radishes at day 0 & day 7

Groups	Day 0	Day 7
T1	3.28 x 10^5 Pa	**2.97 x 10^5 Pa**
T2	4.23 x 10^5 Pa	2.65 x 10^5 Pa
T3	4.48 x 10^5 Pa	2.58 x 10^5 Pa
Control	3.30 x 10^5 Pa	2.55 x 10^5 Pa

Figure 5 shows the bar graph showing the average difference of weight loss in all treated radishes, presented as grams (g):

Figure 5

Firmness of treated radishes at day 7



The group showed the least weight loss, least signs of decay, and greatest firmness was the most effective in prolonging the shelf-life of radish. Based on the results, cabbage extract with 20% concentration was deemed the best concentration that prolonged the shelf-life of radish.

After the researchers determined the most effective concentration, the p-value was calculated of all quantitative parameters using Jamovi version “2.3.28”. The alpha value assigned to the study was 0.05, p-value greater than 0.05 is considered not significant. The statistical analysis used was Welch’s one-way ANOVA and Tukey HSD’s post-hoc test.

Table 4 shows the p-value calculated from all parameters mentioned:

Table 4

One-Way ANOVA p-value of all parameters

Parameters	p-value[a]
Weight Loss	0.915
Signs of Decay	0.958
Firmness	**0.004**

a. $\alpha = 0.05$ ($p < 0.05$ is significant)

Firmness being statistically significant shows a possible interaction between the flavonoids in the cabbage extract and the radish’s firmness. Post-hoc suggests a significance with the 20% concentration. The results suggest a saturation point, which the firmness of the radishes worsen as the concentration exceeds the saturation point.

Radishes treated with the lowest concentration of cabbage extract showed slightly better results in these parameters. The weight loss findings contrasted with Lupu (2024) who reported lower weight loss in radishes stored at room temperature. The post-harvest decay rate of over 60% in all groups also differs from Kumar (2021) who suggests that cabbage extracts reduce bacterial/fungal growth that significantly decreases the signs of decay. Similarly, our results also differed from Lockley (2020) who reported an increase radish stiffness over time at room temperature.

The implications suggest several factors that could potentially affect the results. One factor was the moisture levels in the room, as noted by Huynh (2019) who stated high moisture content contributes to a short radish shelf-life. Which that implication aligns with our observed data. Another factor was the cabbage extract. While the radish and distilled water combination showed a negative decay trend, the extract alone might not directly retain radish moisture, potentially leading to faster decay. Another factor was the temperature of the room. Our storage temperature (30°C) was 5 degrees above room temperature. And another factor was the lack of isolation. Containers with ventilation slits might have allowed pests to intrude and tamper with our data, affecting results.

According to Pateiro (2023), the antioxidant activity associated with phenolic acids and flavonoids of leaf vegetables including cabbages makes them good candidates to replace current synthetic preservatives. The statistical analysis results differed from that implication, as the combination of extrinsic factors and systematic errors that can easily be done might suggest that the cabbage extract is not a current replacement to the current preservatives available.

DISCUSSIONS

Cabbage Extract showed different effects in prolonging the shelf-life of radishes. The cabbage extract with the minimum concentration showed equal weight loss to control, were firmer and showed less decay than control. Radishes treated with higher cabbage extract concentration consistently showed worse results. The cabbage extract shows statistically significant effect in prolonging the shelf-life of radish with a minimum extract used. P-value results for all quantitative parameters are as follows, there was no significant difference for Weight Loss and Signs of Decay, while it shows significant difference for Firmness (T1 vs C and T3 showed the biggest significance).

Based on the findings of the study, the following conclusions are drawn. Firstly, cabbage extract with the least concentration was the most likely to prolong the shelf-life of radish among other groups. Secondly, all radishes treated with cabbage extracts lost more mass, showed less signs of decay, and were firmer than radishes treated with only distilled water. Lastly, all cabbage extract concentrations were not significant in prolonging the shelf-life of radish, except for the least concentrated extract that retained the radishes firmness better than the rest of the applied solutions.

The following recommendations are made for consideration in future studies. Consider different plant sample and extraction methods, choosing plant samples that rot faster that contain lower moisture content and choose different extraction methods that would yield more phytochemicals. Add control without exposure to moisture, adding another set of control without submerging the radishes in water. Explore different storing methods, a properly contained set-up with a constant temperature around room temperature and cover radishes individually are recommended.

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APPENDICES

Appendix A

Cabbage Extract Synthesis

Cycle Count	**20%** (50g Cabbage)	**40%** (100g Cabbage)	**60%** (150g Cabbage)
1	Extract Time: **45 s** Extract Temp: **30°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **28°C** Time Mix: **1:30 s**	Extract Time: **45 s** Extract Temp: **33°C** Time Mix: **1:30 s**
2	Extract Time: **45 s** Extract Temp: **40°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **39°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **61°C** Time Mix: **2:30 s**
3	Extract Time: **45 s** Extract Temp: **50°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **53°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **74°C** Time Mix: **2:30 s**
4	Extract Time: **45 s** Extract Temp: **60°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **58°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **76°C** Time Mix: **2:30 s**
5	Extract Time: **45 s** Extract Temp: **65°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **62°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **74°C** Time Mix: **2:30 s**
6	Extract Time: **45 s** Extract Temp: **70°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **68°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **78°C** Time Mix: **2:30 s**
7	Extract Time: **45 s** Extract Temp: **74°C** Time Mix: **1:15 s**	Extract Time: **45 s** Extract Temp: **73°C** Time Mix: **1:15 s**	

Appendix B

Phytochemical Tests

Alkaline Reagent Test

A.) *NaOH Added*

B.) *HCl Added*

Shinoda Test

A.) *(Mg) Added*

B.) *HCl Added*

Appendix C

Preparation of Cabbage Solutions

From Left to Right: *20% (50g Cabbage), 40% (100g Cabbage), and 60% (150g Cabbage) Concentration with 250mL Distilled Water respectively.*

Appendix D

Initial Measurement of Radishes

Control 1	T1-1	T2-1	T3-1
1 – 105.5 g 2 – 105.0 g 3 – 107.0 g 4 – 121.0 g	1 – 118.5 g 2 – 115.5 g 3 – 107.5 g 4 – 112.0 g	1 – 121.5 g 2 – 107.0 g 3 – 129.0 g 4 – 127.5 g	1 – 110.0 g 2 – 128.5 g 3 – 113.0 g 4 – 130.5 g
Control 2	**T1-2**	**T2-2**	**T3-2**
1 – 88.0 g 2 – 112.0 g 3 – 139.5 g 4 – 117.0 g	1 – 104.5 g 2 – 97.5 g 3 – 104.0 g 4 – 106.5 g	1 – 137.5 g 2 – 131.0 g 3 – 131.0 g 4 – 133.5 g	1 – 85.5 g 2 – 97.5 g 3 – 99.5 g 4 – 111.5 g
Control 3	**T1-3**	**T2-3**	**T3-3**
1 – 146.0 g 2 – 112.0 g 3 – 139.5 g 4 – 170.0 g	1 – 147.0 g 2 – 152.0 g 3 – 132.5 g 4 – 137.0 g	1 – 124.5 g 2 – 130.0 g 3 – 127.5 g 4 – 134.5 g	1 – 168.0 g 2 – 145.5 g 3 – 140.5 g 4 – 155.5 g
Control 4	**T1-4**	**T2-4**	**T3-4**
1 – 236.0 g – $3x10^6$ Pa 2 – 301.5 g – $3x10^5$ Pa 3 – 232.0 g – $3.5x10^5$ Pa 4 – 230.5 g – $3.7x10^5$ Pa	1 – 189.5 g – $3.5x10^6$ Pa 2 – 219.0 g – $3x10^5$ Pa 3 – 191.5 g – $3.2x10^6$ Pa 4 – 141.5 g – $3.4x10^6$Pa	1 – 153.0 g – $3.8x10^6$ Pa 2 – 176.5 g – $4.8x10^5$ Pa 3 – 199.5 g – $4.8x10^5$ Pa 4 – 187.5 g – $3.5x10^5$ Pa	1 – 195.5 g – $4.8x10^5$ Pa 2 – 305.5 g – $4x10^5$ Pa 3 – 170.0 g – $4.3x10^5$ Pa 4 – 264.0 g – $4.8x10^5$ Pa

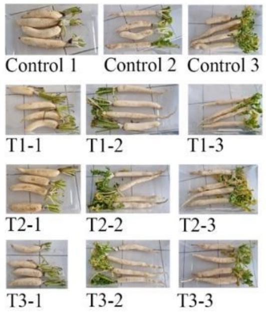
Appendix E

Final Measurement of Radishes

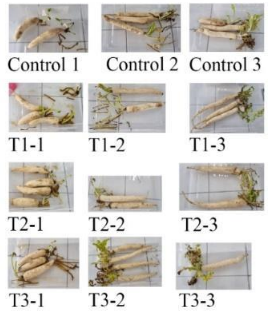
Control 1	T1-1	T2-1	T3-1
1 – 88.0 g 2 – 92.5 g 3 – 94.0 g 4 – 99.5 g	1 – 103.0 g 2 – 106.5 g 3 – 102.0 g 4 – 107.0 g	1 – 107.0 g 2 – 94.5 g 3 – 113.5 g 4 –114.5 g	1 – 99.0 g 2 – 113.0 g 3 – 104.5 g 4 – 118.0 g
Control 2	**T1-2**	**T2-2**	**T3-2**
1 – 75.5 g 2 – 100.0 g 3 – 126.0 g 4 – 99.0 g	1 – 93.5 g 2 – 84.5 g 3 – 94.0 g 4 – 94.5 g	1 – 115.0 g 2 – 107.5 g 3 – 120.5 g 4 – 115.5 g	1 – 77.5 g 2 – 80.5 g 3 – 83.5 g 4 – 99.5 g
Control 3	**T1-3**	**T2-3**	**T3-3**
1 – 137.5 g 2 – 144.5 g 3 – 134.0 g 4 – 149.0 g	1 – 134.0 g 2 – 138.0 g 3 – 120.5 g 4 – 124.0 g	1 – 108.5 g 2 – 117.0 g 3 – 116.0 g 4 – 125.5 g	1 – 145.5 g 2 – 119.0 g 3 – 122.0 g 4 – 129.0 g
Control *(Penetrometer)*	**T1** *(Penetrometer)*	**T2** *(Penetrometer)*	**T3** *(Penetrometer)*
C1, 1 – $2.41x10^5$ Pa C1, 2 – $2.31x10^5$ Pa C2, 3 – $2.29x10^5$ Pa C3, 2 – $2.52x10^5$ Pa	T1-3, 3 – $2.71x10^5$ Pa T1-3, 2 – $2.81x10^5$ Pa T1-2, 2 – $2.62x10^5$ Pa T1-2, 1 – $2.90x10^5$ Pa	T2-3, 2 – $2.51x10^5$ Pa T2-3, 3 – $2.83x10^5$ Pa T2-2, 2 – $2.69x10^5$ Pa T2-2, 4 – $2.59x10^5$ Pa	T3-3, 2 – $2.82x10^5$ Pa T3-3, 1 – $2.78x10^5$ Pa T3-3, 4 – $2.60x10^5$ Pa T3-1, 1 – $3x10^5$ Pa

Appendix F

All Radishes in Day 0 and Day 7



Day 0



Day 7

Appendix G

Thermometer/Hygrometer readings in LSB-711 for the 7-day observation period

Parameter	High	Low
Temperature[a]	30.4°C	15.6°C
Humidity	75%	32%

a. room temperature is 25°C.

Appendix H

Welsch's one-way ANOVA and Tukey HSD post-hoc result

ANOVA Results (12STEM1_G8)

Weight Loss One-Way ANOVA

One-Way ANOVA (Welch's)

	F	df1	df2	p
Weight Loss	0.171	3	24.3	0.915

Group Descriptives

	WL-CG	N	Mean	SD	SE
Weight Loss	Control	12	59.1	43.3	12.5
	T1	12	59.0	40.4	11.7
	T2	12	69.2	46.2	13.3
	T3	12	68.7	58.6	16.9

Firmness One-Way ANOVA

One-Way ANOVA (Welch's)

	F	df1	df2	p
Firmness	15.2	3	5.83	0.004

Group Descriptives

	F-CG	N	Mean	SD	SE
Firmness	Control	4	255000	20817	10408
	T1	4	297500	9574	4787
	T2	4	265000	23805	11902
	T3	4	257500	5000	2500

Post Hoc Tests

Tukey Post-Hoc Test – Firmness

		Control	T1	T2	T3
Control	Mean difference	—	-42500 *	-10000	-2500
	p-value	—	0.017	0.832	0.996
T1	Mean difference		—	32500	40000 *
	p-value		—	0.073	0.024
T2	Mean difference			—	7500
	p-value			—	0.919
T3	Mean difference				—
	p-value				—

Note. * p < .05, ** p < .01, *** p < .001

Signs of Decay One-Way ANOVA

One-Way ANOVA (Welch's)

	F	df1	df2	p
Signs of Decay	0.0977	3	4.40	0.958

Group Descriptives

	SOD-CG	N	Mean	SD	SE
Signs of Decay	Control	3	83.3	28.9	16.7
	T1	3	66.7	38.2	22.0
	T2	3	75.0	25.0	14.4
	T3	3	75.0	25.0	14.4